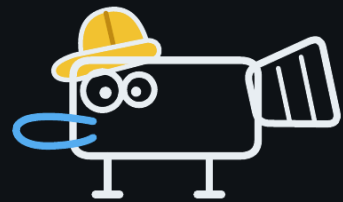


MURDER BOARD · ROUND 1 · FOR PRATEEK

Your questions, answered.

Sir, thank you for the questions. Each section below takes one of them in three parts: **what you asked**, **how I answer it**, and **the numbers that justify it**.

- | | |
|----|----------------------------|
| 01 | Proto is the moat |
| 02 | Softer claims |
| 03 | One product at a time |
| 04 | Competition and market |
| 05 | Automation, honestly |
| 06 | Competitors' costing |
| 07 | What it costs, and payback |
| 08 | Scale |
| → | Where this is going |



proto plans · [Baker](#) makes · [Mixer](#) tests · [Probe Matrix](#) reaches every net
Soham, founder · 2 October 2026 · [protopus-kappa.vercel.app](#)

BEFORE WE START

Sir, your feedback changed the pitch. Here is what is different now.

PROTO FIRST

Proto is the company. [Baker](#), [Mixer](#) and the [Probe Matrix](#) are its hands and eyes in the real world.

PROTOTYPING, NOT PRODUCTION

[Baker](#) is for the first 1–10 boards of a design. Volume still goes to a fab.

SOFTER CLAIMS

Lower cost per try and shorter lead time, with every number marked projected.

BAKER FIRST

₹15 lakh for nine months, or ₹10 lakh lean. [Mixer](#) and the [Probe Matrix](#) wait for step 2.

PAYBACK

[Baker](#) pays for itself after 47 boards against JLCPCB: about six months for a startup.

HONEST SCALE

One [Baker](#) makes 750–1,250 boards a year. Beyond that, a fab is the right tool.

ABOUT THE NUMBERS

Prices are projected: [Baker](#) ₹2,99,000, [Mixer](#) ₹34,999 (step 2), Proto ₹9,999 a year. Cycle times are engineering targets, and the proof of concept will measure them. The numbers come from PROTOPUS_Economics_Model.xlsx and PROTOPUS_Baker_Mixer_BOM_and_Costing.xlsx (2 Oct 2026). USD is converted at ₹88.

QUESTION 01 · PROTO IS THE MOAT

The moat is Proto. The machines are its hands and eyes.

SIR, YOU ASKED

“The strongest moat is hidden in the idea. If I were the founder, I would pitch Proto instead of two hardware robots, [Baker](#) and [Mixer](#). And we should not project this as a desktop PCB mill, which has a rough history.”

I CAN ANSWER

You’re right, sir. Proto is the company. But an AI can only build hardware if it has hands and eyes in the real world.

Today a commerce graduate can vibe-code an app, because the model reaches every tool it needs through APIs and connectors. Hardware has no API. A model can already draw a board (Flux.ai produces Gerbers), and Claude can drive Fusion 360 and Blender. But no model can etch copper, place a part, or tell whether the I²C bus on a real board is alive. [Baker](#) and [Mixer](#) are that missing API, and the [Probe Matrix](#) gives Proto a finger on every net, so it can test and program the board.

And [Baker](#) isn’t a desktop mill. A mill automates one step for people who are already experts. [Baker](#) runs fifteen steps in one machine, with Proto guiding a first-timer through each one.

TO JUSTIFY



AI today: Flux gives Gerbers, Claude drives Fusion and Blender. It stops at the file.

protopus: the AI's hands and eyes

THE HANDS

[Baker](#)

makes the board: mill, drill, mask, stencil, paste, reflow

THE EYES

[Mixer](#)

signal generator and analysers; replays real sensor faults (step 2)

THE FINGERS

[Probe Matrix](#)

probes every net, so Proto can test and program the board

THE REAL MOAT IS THE DATA

Every run logs design → machine settings → measurement → outcome. A tool that only produces files never sees the real board, so it can never collect this data. The longer Protopus runs, the better Proto gets.

NOT A DESKTOP PCB MILL

A mill automates one step for experts, and that category has a rough history: Bantam Tools stopped selling its PCB mill in 2025. [Baker](#) runs 15 steps in one coordinate frame, from bare FR4 to a reflowed board, and Proto plans each step.

QUESTION 01 · TO JUSTIFY, CONTINUED

How people learn hardware, and why ease is the market

People learn practical skills by doing: try, look, adjust (Kolb’s experiential learning cycle). The speed of that loop sets the speed of learning. Hardware makes a beginner climb eight rungs, and each rung loses people. Proto takes rungs away, and each rung removed brings in a new crowd: students, designers, domain experts and first-time founders.

RUNG	YOU FIRST NEED TO KNOW	WHERE PEOPLE GET STUCK	WHO CARRIES IT NOW
1 · idea → spec	how to state requirements	vague goals, scope creep	Proto asks, then writes the spec
2 · parts	datasheets, footprints, stock	hours of searching; stock runs out	Proto picks in-stock parts
3 · schematic	circuit theory, references	errors that only show on the bench	Proto + Flux check it
4 · layout	an EDA tool, DRC	weeks to learn the tool	Proto fixes it (Baker rules)
5 · fabrication	Gerbers, vendors, customs	7-20 days to India per try	Baker, in hours (target)
6 · assembly	SMD soldering, stencils, reflow	skill and equipment	Baker + a printed jig
7 · bring-up	scope, logic analyser, firmware	you can’t tell what’s wrong	bench now; Mixer + Probe Matrix in step 2
8 · iterate	root-cause debugging, DFM	slow loops, fewer attempts	Proto explains the fix

7–20 days

per try today: JLCPCB to India, door to door

≈2 hours

per try on Baker (target)

\$100M ARR

Lovable, 8 months after removing the coding rung

Rungs 1–4 are already being solved in software. Rungs 5–8 are physical, which is why AI-only tools stop at the file, and why Proto needs machines.

Sources: Kolb (1984) · TechCrunch, 23 Jul 2025 (Lovable) · zbotic.in (JLCPCB to India).

QUESTION 02 · SOFTER CLAIMS

We don't replace the fab. We fill the gap before it.

SIR, YOU ASKED

“Avoid strong claims like “replaces a fabrication order”. An investor who has invested in hardware manufacturing companies: how would he take it? Softer claims are better, like lower cost, faster lead time, etc.”

I CAN ANSWER

Fair point, sir, and we have changed it. We never meant to replace fab houses. Protopus fills the gap between a finished design and a board you can actually test in the real world.

Think of a plastic bottle. You 3D-print it first to check its shape, feel and fit, and you pay for the injection mould only for the first thousand. Plastic has had that desktop prototyping path for years. Electronics never got a desktop path that ends in a finished, soldered, tested board. [Baker](#) is that path. Once a design is frozen, Proto sends the fab files that have already worked on a real board. **So an investor in manufacturing gets a company that sits upstream of the fabs and sends them better first orders.**

TO JUSTIFY

WE USED TO SAY	NOW WE SAY	BASIS
“replaces a fabrication order”	lower cost per try and same-day lead time, for the first 1-10 boards of a design	₹334 of materials vs ₹6,721 from JLCPCB landed in India, 7-20 days
“send it at night, pick it up in the morning”	about 2 hours per board, 15-20 minutes of it hands-on (target)	cycle-time model: 81-141 minutes
“fully automatic”	steps 1-12 automatic; you place the parts and start the reflow	see question 5
“replaces your bench”	Analog Discovery 3-class testing, run by Proto (step 2)	Mixer production cost ₹13,740 (projected)

THE PATH FROM IDEA TO 1,000S

	PROTOTYPE	VOLUME
a plastic bottle	desktop 3D print: shape, feel, fit	injection mould
a circuit board, today	fab order + hand assembly + a borrowed bench	fab / EMS
a circuit board, with us	Proto + Baker : a finished board on the desk	fab / EMS, with files already proven on a real board

Every motor speed, processing time and jig-print time in our material is a projected engineering estimate. The proof of concept exists to measure them.

QUESTION 03 · ONE PRODUCT AT A TIME

Baker first. The rest of the vision comes step by step.

SIR, YOU ASKED

“Two products for a first-shot raise of ₹10L is, in my opinion, a bit aggressive. We should go step by step. Maybe start with Baker first?”

I CAN ANSWER

Agreed, sir. We will go one product at a time, starting with Baker and Proto.

The two machines are the vision: one makes the hardware, the other tests it, programs it and feeds the results back to Proto. But the money we raise now builds only Baker. While we prove it, its boards are tested on ordinary bench instruments. Mixer and the Probe Matrix come in step 2, funded by the seed round after the proof of concept.

TO JUSTIFY

STEP 1 · NOW · ₹15L

Proto + Baker

all 15 steps running, 20 proof-of-concept boards, cycle time measured, Proto from brief to Baker job

STEP 2 · SEED · ~₹1.5 CR

Mixer + Probe Matrix

test and program every board; pilots of 20 Bakers and 100 Mixers built turnkey in China

STEP 3 · ECOSYSTEM

Proto runs it all

plus any 3D printer you already own; India first, then global

WHERE ₹15,00,000 GOES · 9 MONTHS · NO FOUNDER SALARY

LINE	₹
<u>Baker</u> build, costed line by line (70-line BOM, duty, 15% contingency)	3,73,418
<u>Baker</u> Rev B + ~150 test boards of materials	1,60,000
one software engineer who also writes firmware, 9 × ₹50,000	4,50,000
Proto software + compute, 9 × ₹15,000	1,35,000
testing, tooling, lab safety	1,50,000
IP + legal, outreach, reserve	2,31,582
Total	15,00,000

IF THE ROUND IS CAPPED AT ₹10L

Baker ₹3.73L + Rev B ₹1.0L + the engineer for 6 months ₹3.0L + compute ₹0.9L + testing ₹1.0L = ₹9.63L, leaving ₹0.37L in reserve. This version has no lab set-up, IP or outreach, and fundraising would have to start in month 4.

QUESTION 04 · COMPETITION AND MARKET

Everyone else solves one step. We run the whole loop.

SIR, YOU ASKED

“I think a thorough competition analysis and market study is needed.”

I CAN ANSWER

We did the study, sir, and here it is in one paragraph.

Protopus sits where two large markets meet: printed circuit boards (₹6.4 lakh crore worldwide; ₹55,000 crore in India, growing about 15% a year) and test and measurement (₹3.1 lakh crore worldwide; ₹8,300 crore in India). Our first market is much smaller, and we can count it. India has 5,868 AICTE engineering colleges and about 2,250 hardware startups. That is 8,118 buyers, or about ₹243 crore of [Baker](#) sales at one machine each, plus ₹8.1 crore a year in Proto plans. Behind them come 10,000 Atal Tinkering Labs, with 50,000 more announced. **Every competitor solves one step.** A mill makes a bare board for an expert. A placer only places parts. A USB instrument measures, but a human has to read the screen. A fab ships in weeks, and an AI design tool stops at the file. **Nobody runs the whole loop, and nobody lets an AI drive the machines. That is where we are different.**

TO JUSTIFY

₹6.4L cr global PCB market	₹3.1L cr global test + measurement	₹55K cr India PCB, ~15% a year	₹8.3K cr India test + measurement
INDIA BEACHHEAD	BUYERS	BAKER (₹2,99,000)	WITH MIXER (₹3,33,999)
AICTE engineering colleges	5,868	₹175 Cr	₹196 Cr
hardware startups (1% of 2.25 lakh DPIIT)	2,250	₹67 Cr	₹75 Cr
Total, one machine each	8,118	₹243 Cr	₹271 Cr

Plus ₹8.1 Cr a year of Proto plans (₹9,999 × 8,118). Next come 10,000 Atal Tinkering Labs, each with ₹20 lakh over five years and 50,000 more planned, then ~2,500 Fab Labs worldwide, 33M Arduino users and 2.3M Hackster members.

	MAKE	ASSEMBLE	TEST	AI GUIDE
Desktop mills (Makera Carvera, LPKF)	●	○	○	○
Ink printer (Voltera V-One)	●	●	○	○
Pick-and-place (Neoden YY1)	○	●	○	○
USB instruments (Digilent AD3, Moku:Go)	○	○	●	○
Fabs (JLCPCB, PCBWay)	●	●	○	○
AI design (Flux.ai)	○	○	○	●
protopus: Proto + Baker (+ Mixer)	●	●	●	●

Sources: IMARC · Grand View · AICTE · DPIIT/TiE (2.25 lakh startups; the 1% hardware share is our assumption) · Atal Innovation Mission · Union Budget 2025-26 · Fab Foundation · Arduino · Hackster.

QUESTION 05 · AUTOMATION, HONESTLY

Not fully automatic. Same-day, with about fifteen minutes of your hands.

SIR, YOU ASKED

“Send the design at night, pick it up in the morning’ is a bit too aggressive a claim. We have to load boards, place components, etc. So it’s not fully automated.”

I CAN ANSWER

You’re right, sir, it isn’t fully automatic, and we no longer say it is.

Boards don’t need loading every time, because a refill magazine feeds the next blank and you only top it up. The placement jig is generated automatically from the Gerber and printed in parallel with the board. Reflow and jig-making happen inside the machine, but not hands-free: you drop the parts into the jig and set it on the reflow plate. So the honest promise is: **start a board after lunch, and test it the same afternoon.**

TO JUSTIFY

STEP	WHAT HAPPENS	WHO	MINUTES
01-02	feed the blank from the refill magazine, probe the height map	machine	5
03	isolation mill + drill	machine	30-55
04-09	clean, coat the mask, UV image, develop, cure	machine	16-27
10-12	laser-cut the stencil, clamp, paste, release	machine	8-13
jig	generated from the Gerber, printed in parallel with steps 01-12	machine	+0
13	you drop ~40 parts into the jig with a vacuum pen	you	7-14
14	you set the jig on the plate; the 138 °C reflow profile runs itself	you + machine	10-12
15	test: Proto runs the plan (bench now; Probe Matrix in step 2)	you hook up	5-15
total	per board, one operator	15-20 min hands-on	81-141

81-141 min

per board (target)

15-20 min

of it hands-on

3-5

boards per 8-hour shift

750-1,250

boards a year, one shift

Source: Throughput & Scale sheet, Economics Model. These are engineering targets for the first prototype; the proof of concept measures them.

QUESTION 06 · COMPETITORS' COSTING

The same loop costs ₹7.8 lakh elsewhere, and still has no AI.

SIR, YOU ASKED

"I also strongly recommend, as mentioned above, looking at the competition and their costing."

I CAN ANSWER

Here is what the same loop costs elsewhere, sir.

Bought as separate machines (a Makera Carvera mill, a Neoden YY1 placer and a Digilent Analog Discovery 3) it comes to about ₹7.82 lakh before import duty. That set-up still has no solder mask, no stencil, and no AI tying it together. [Baker](#) with Proto is ₹2.99 lakh, and adding [Mixer](#) brings it to ₹3.34 lakh. On our side, [Baker](#)'s production cost is ₹1,44,842, a 51.6% gross margin, and [Mixer](#)'s is ₹13,740, a 60.7% gross margin.

TO JUSTIFY

PRODUCT	WHAT IT DOES	LIST PRICE	WHERE IT LOSES TO US
LPKF ProtoMat	professional PCB mill	₹8.6–24.5L per machine	separate machines for mask, plating, assembly; for experts
Voltera V-One	prints conductive ink; dispenses paste	~₹3.1L (\$3,500)	ink traces, not etched copper; no mask, stencil or test
Makera Carvera / Air	desktop CNC mill	₹4.8L / ₹2.2L	mills only
Bantam Tools PCB mill	desktop PCB mill	discontinued 2025	the "rough history": one step, for experts
Neoden YY1	pick-and-place	~₹2.6L	places parts only
Digilent AD3	USB scope + logic + supplies	~₹34,000	a human reads the screen; it doesn't know the design
Moku:Go / Red Pitaya / ADALM2000	USB or FPGA instruments	₹21,000–70,000	measurement without the design or a verdict
JLPCB / PCBWay	fab + assembly	₹6,721 per try, landed	7–20 days; minimum orders; customs
Flux.ai	AI PCB design (SaaS)	₹1,300–4,300 / editor / month	stops at the file; Proto uses its API and goes on
protopus: Proto + Baker	board, mask, stencil, paste, reflow; AI-planned	₹2,99,000 + ₹9,999/yr	Mixer ₹34,999 + Probe Matrix in step 2

THE SAME LOOP, BOUGHT SEPARATELY VS PROTOPUS

mill + placer + AD3		₹7.82L: no mask, stencil or AI
Baker + Mixer		₹3.34L: the whole loop
Proto + Baker		₹2.99L: step 1

List prices at ₹88 per US\$, before Indian duty and GST. Sources: LPKF via Design World · Voltera · Makera · ToolGuyd · EEVblog · RS/RobotShop · Liquid Instruments · Red Pitaya · Mouser · JLPCB + zbotic.in · flux.ai. Protopus prices projected.

QUESTION 07 · WHAT IT COSTS, AND PAYBACK

₹2,99,000. It pays for itself after 47 boards.

SIR, YOU ASKED

“How much is this machine going to cost? Say X lacs: we also need to see how many boards are created and whether it cancels my cost.”

I CAN ANSWER

Baker will be ₹2,99,000, sir (projected, before GST).

On Baker, a board costs about ₹334 in materials: copper-clad board, mask, paste, stencil film, jig resin, bit wear and electricity. The same assembled board from JLCPCB, landed in India with shipping and customs, costs ₹6,721 and takes 7–20 days. So Baker cancels its cost after **47 boards**. That’s about six months for a startup doing two boards a week, and about a year for a college lab doing one.

And this count is conservative. It leaves out the engineer’s time and the days spent waiting, which are usually worth more than the boards.

TO JUSTIFY

₹2,99,000		₹334	₹6,721	47 boards	
Baker list price (projected)		materials per board on Baker	same board, JLCPCB landed in India	to pay Baker back	
CUSTOMER BUYS	COMPARED WITH	SAVING PER BOARD	BOARDS TO PAY BACK	STARTUP, 2 A WEEK	LAB, 1 A WEEK
Baker	JLCPCB to India	₹6,387	47	5.9 months	11.8 months
Baker	Indian 24-h fab	₹5,166	58	7.2 months	14.5 months
Baker + Mixer	JLCPCB to India	₹6,387	53	6.6 months	13.2 months
IF THE BAKER PRICE MOVES		₹2,99,000	₹3,49,000	₹3,99,000	₹4,49,000
boards to pay back Baker		47	55	63	71
boards to pay back Baker + Mixer		53	61	68	76
our gross margin on Baker		51.6%	58.5%	63.7%	67.7%

FROM OUR SIDE

Each Baker earns ₹1,54,158 of gross profit (51.6%), or about ₹99,000 after all costs. So 10 Bakers sold return the whole ₹15 lakh round on gross profit, and 16 do so on all-in profit.

Startup = 2 boards a week; lab = 1 a week. JLCPCB: 5 PCBs + PCBA (\$8 setup, \$1.50 stencil) + DHL + 10% BCD + 18% IGST. Adding the first year of the Proto plan (₹9,999) moves Baker’s payback to 49 boards.

QUESTION 08 · SCALE

Baker scales the number of people who can prototype, not boards per machine.

SIR, YOU ASKED

“Innovation is good but may not be scalable in a production environment. 1,000 boards? 10,000? 1 lakh?”

I CAN ANSWER

Baker isn't meant for production, sir, and we won't claim it is. It is for the first one to ten boards of a design.

At 10,000 or a lakh boards, a fab is the right tool, and Proto hands it files that have already worked on a real board. What scales is the number of people who can prototype. Proto is software, so each extra user costs only compute. Our own manufacturing scales too: it is turnkey in China, with final test, minimal assembly and shipping from India.

TO JUSTIFY

BOARDS A YEAR	BAKERS NEEDED (1 SHIFT)	RIGHT TOOL	WHY
10	1	<u>Baker</u>	same-day loops beat fab minimums and shipping
100	1	<u>Baker</u>	one machine covers a busy lab or startup
1,000	1-2	<u>Baker</u> + a fab	iterations on <u>Baker</u> ; any batch of 10+ identical boards goes to a fab
10,000	14	fab / EMS	production; Proto exports the checked Gerber, BOM and placement files
1,00,000	134	fab / EMS	not what <u>Baker</u> is for, and we don't claim it

HOW WE BUILD

China, turnkey

parts sourced and assembled in China, then factory-tested

HOW WE SHIP

India

final test, minimal assembly and shipping; India first, then global

HOW PROTO SCALES

Software

each extra user costs only compute; every board makes it better

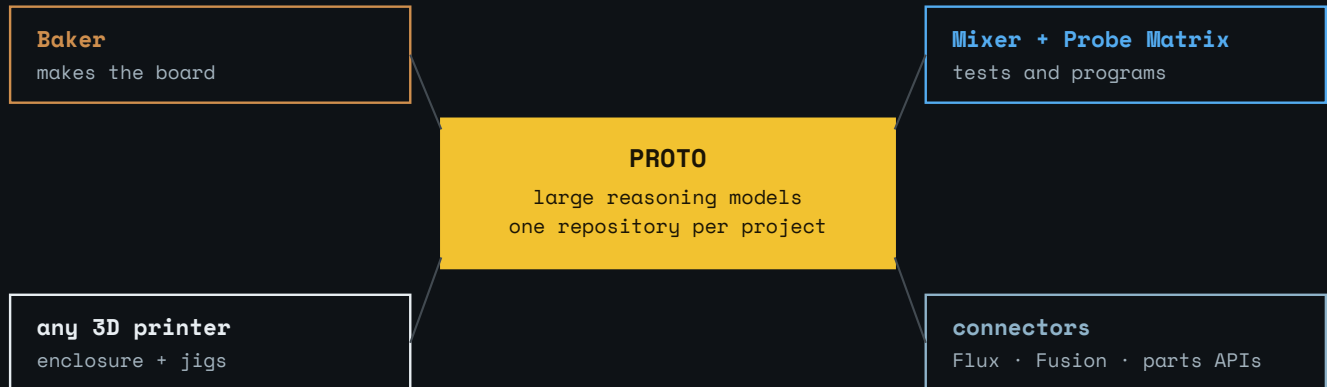
THE NUMBERS BEHIND IT

One Baker makes 750–1,250 boards a year (3–5 per shift, target). Production cost is ₹1,44,842 per Baker at 100 units, with RFQs planned from 5 to 5,000 units. The China master plan covers the plant, the suppliers and the quality gates.

AND WHERE THIS IS GOING

One plain-English brief in. One tested prototype out.

Sir, this is the vision. Proto runs [Baker](#), which makes the board. It runs [Mixer](#) with the [Probe Matrix](#), which tests and programs it. And it runs **any 3D printer you already own**; Protopus will never sell a printer.



PROTO'S BRAIN

Large reasoning models break the goal into steps and call tools: EDA through the Flux.ai API, CAD through Fusion 360 and Blender, parts suppliers, and our machines.

ELECTRONICS

A BOM of in-stock parts, each with a purchase link. Affiliate links are a possible extra revenue line.

MECHANICAL

A 3D-printable enclosure sized from the finished board: heights from the jig, outline and holes from the Gerber.

EXAMPLE · A SMARTWATCH WITH 10 PHYSICAL BUTTONS

#	WHAT HAPPENS
01	You describe it in plain English: "a smartwatch with 10 physical buttons, a heart-rate sensor, a week of battery".
02	Proto checks feasibility and limits (size, battery, buttons against wrist space) and proposes the best prototype for the goal.
03	Proto writes the requirements back in plain English for you to approve.
04	Proto designs the electronics (MCU, sensors, buttons, power) and produces the PCB and a BOM with purchase links.
05	Baker makes the board, and your 3D printer prints the case, sized from the measured board.
06	Proto writes and flashes the firmware through Mixer and the Probe Matrix .
07	Mixer tests every function with real signals, flags failing parts or nets, and gives a plan of action.
08	You wear it and tell Proto what felt off ("buttons too stiff, screen too dim").
09	Proto plans the next revision and the loop runs again. When you're ready for volume, the final files go to a fab.

WHO WE ARE

Protopus builds smart prototyping solutions: smart machines and accessories, run by an AI engineer. [Baker](#) is the first product, with [Mixer](#) and the [Probe Matrix](#) next. Software and hardware teams keep improving both. Manufacturing is turnkey in China, with testing, minimal assembly and shipping from India, to India first and then the world.

Thank you, sir.

Happy to get on a call and go through any of this in detail.

[website](#) · [Baker](#) · [Mixer](#) · [Probe Matrix](#)

Soham · Founder, PROTOPUS · soham9763015@ece.du.ac.in

